

Activity 20 Instruction Manual

Activity 20 | Topic 03 | C013 | Version v1.0

Purpose

When a decision carries real money, qualitative scoring is not enough. You will run a sensitivity analysis and a decision tree, using GenAI for the setup and doing the arithmetic yourself.

Files used

- activity-guide.md and activity-guide.pdf - brief and source instructions
- prompts.pdf - reusable prompt and verification checklist
- mock-data.xlsx - safe fictional project data for practice

Expected output

A tornado chart identifying your most sensitive risk variables and a decision tree with expected monetary values.

Before you begin

1. Form the required group and assign one person to capture evidence.
2. Open mock-data.xlsx and review the title, note and column headings.
3. Open prompts.pdf beside your chosen GenAI tool. Do not upload confidential workplace data.

Detailed procedure

1. In your group, open 'Tornado Chart - Sensitivity Analysis.xlsx'.
2. Prompt GenAI: 'List 10 risk variables that could affect the cost or schedule of a digital transformation project. Estimate high and low values for each.'
3. Enter each variable with its high/low range and plot the bars, longest at the top. The variables at the top of the tornado are where your management attention belongs.
4. Sanity-check the ranges. GenAI produces symmetric ranges by default; real project risks are usually asymmetric - things go wrong further than they go right. Widen the downside where your experience says so.
5. Open 'Decision Tree Analysis.xlsx'. Define a real either/or decision from your project - build vs buy, or two vendors.
6. Prompt GenAI: 'Structure this decision as a decision tree. For each branch give the cost, the possible outcomes with their probabilities, and the payoff of each outcome.'
7. Calculate the Expected Monetary Value for each branch yourself: $EMV = \text{sum of (probability} \times \text{payoff) for each outcome, minus the branch cost}$. Do NOT accept the AI's arithmetic - verify every multiplication.
8. Compare your computed EMV with GenAI's. Where they differ, find the error. This is the single most important habit when using AI for quantitative work.
9. State your recommended decision and the EMV that supports it. Share with the class.

Use the supplied mock data

Copy only the rows needed for the current step and tell the model the data is fictional. If the model changes an identifier or number, reject the change and rerun with an explicit preserve-values instruction.

Quality control

- Your tornado chart is ordered by sensitivity with justified asymmetric ranges, and you have hand-verified the EMV of every decision tree branch.
- Compare the first AI draft with the final human-corrected artifact and retain both as evidence.
- Label assumptions, decisions, owners and dates so another project manager can audit the result.

Troubleshooting

If the response is generic, add the project objective, constraints, audience and required columns. If arithmetic is inconsistent, request formulas and substitutions, then calculate independently. If a source cannot be opened, remove the claim or replace it with verified evidence.

Submission evidence

Submit the completed project artifact, the final prompt, a short critique of the AI output, and any screenshot or exported file required by the activity.